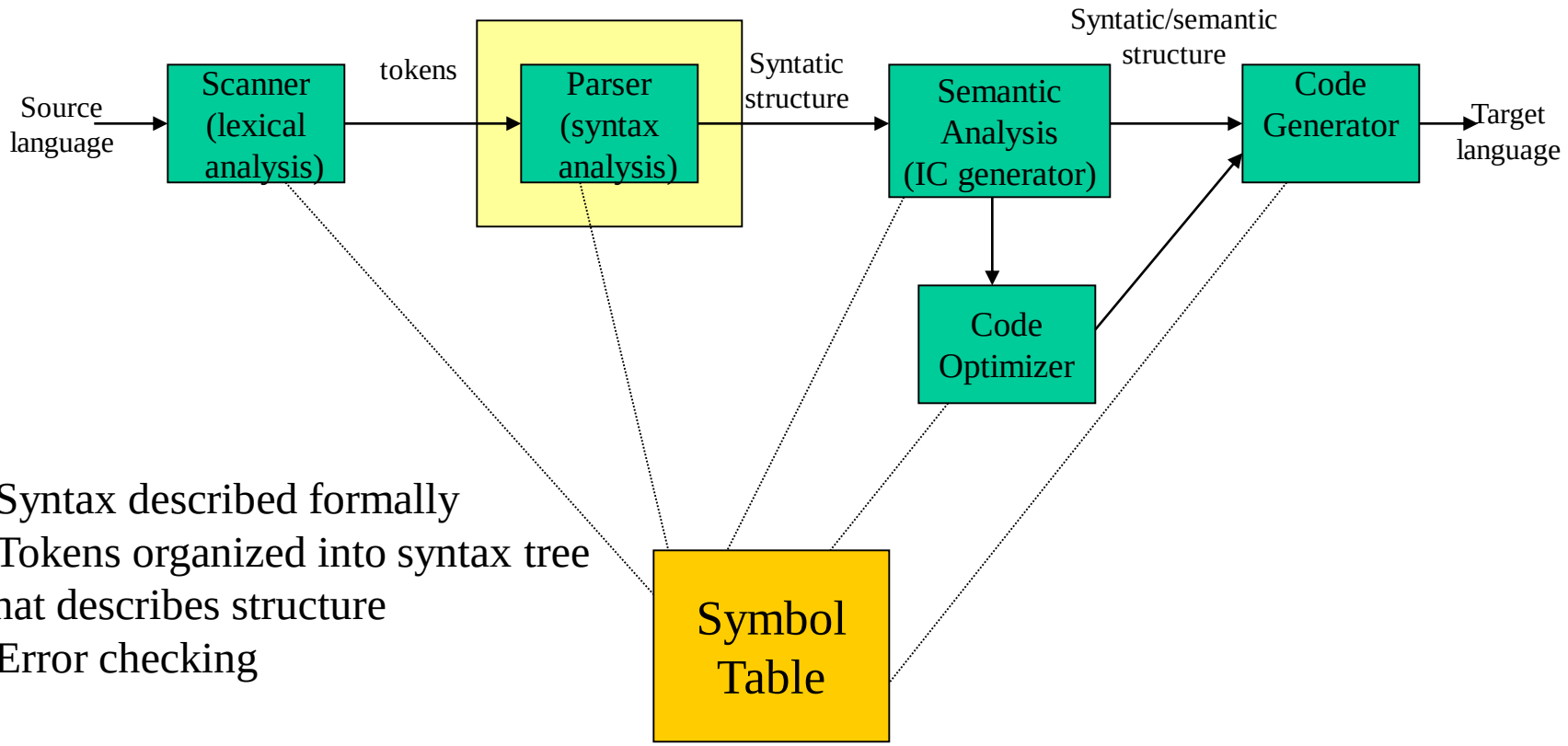
Lecture 3: Parsing

Compiler Design

2nd 2019-2020

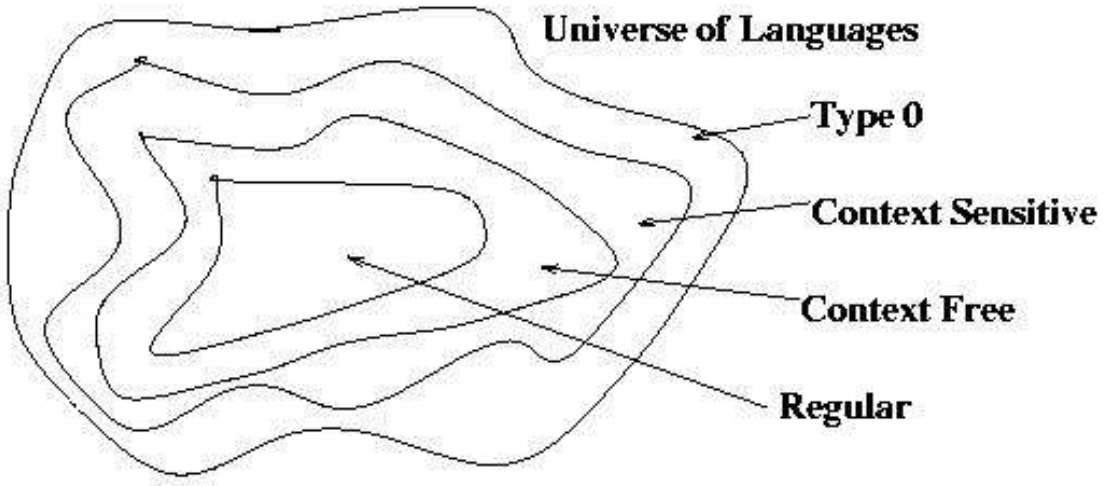
Static Analysis - Parsing



- Syntax described formally
- Tokens organized into syntax tree that describes structure
- Error checking

Static Analysis - Parsing

We can use context free grammars to specify the syntax of programming languages.



Context Free Grammars

Definition: A **context free grammar** is a formal model that consists of:

1. A set of **tokens** or terminal symbols V_t
2. A set of nonterminal symbols V_n
3. Start symbol S (in V_n)
4. Finite set of productions of form:

   $A \rightarrow R_1 \ldots R_n$ ($n >= 0$) where A in V_n,

   R in $V_n \cup V_t$

CFG Examples

Indicates a production

$V_t = \{+,-,0..9\}, \; V_n = \{L,D\}, \; s = \{L\}$

$L \rightarrow L + D \mid L - D \mid D$

$D \rightarrow 0 \mid \ldots \mid 9$

Shorthand for multiple productions

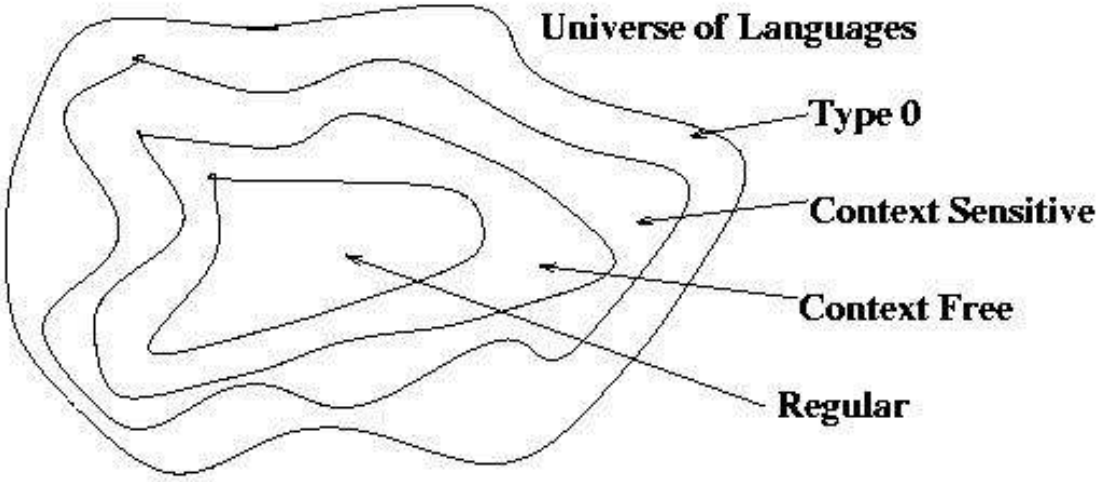
$V_t = \{(,)\}, \; V_n = \{L\}, \; s = \{L\}$

$L \rightarrow (\; L \;) \; L$

$L \rightarrow \varepsilon$

Languages

Regular	$A \rightarrow a\ B,\ C \rightarrow \varepsilon$
Context free	$A \rightarrow \alpha$
Context sensitive	$\alpha A \beta \rightarrow \alpha \gamma \beta$
Type 0	$\alpha \rightarrow \beta$



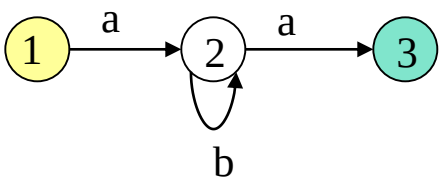
Any regular language can be expressed using a CFG

Starting with a NFA:

- For each state S_i in the NFA
  - Create non-terminal A_i
  - If transition $(S_i,a) = S_k$, create production $A_i \rightarrow a\ A_k$
  - If transition $(S_i,\varepsilon) = S_k$, create production $A_i \rightarrow A_k$
  - If S_i is a final state, create production $A_i \rightarrow \varepsilon$
  - If S_i is the NFA start state, $s = A_i$
- *What does the existence of this algorithm tell us about the relationship between regular and context free languages?*

NFA to CFG Example

ab*a



$A_1 \rightarrow a\ A_2$

$A_2 \rightarrow b\ A_2$

$A_2 \rightarrow a\ A_3$

$A_3 \rightarrow \varepsilon$

Writing Grammars

When writing a grammar (or RE) for some language, the following must be true:

1. All strings generated are in the language.
2. Your grammar produces all strings in the language.

Try these:

- Integers divisible by 2
- Legal postfix expressions
- Floating point numbers with no extra zeros
- Strings of 0,1 where there are more 0 than 1

Parsing

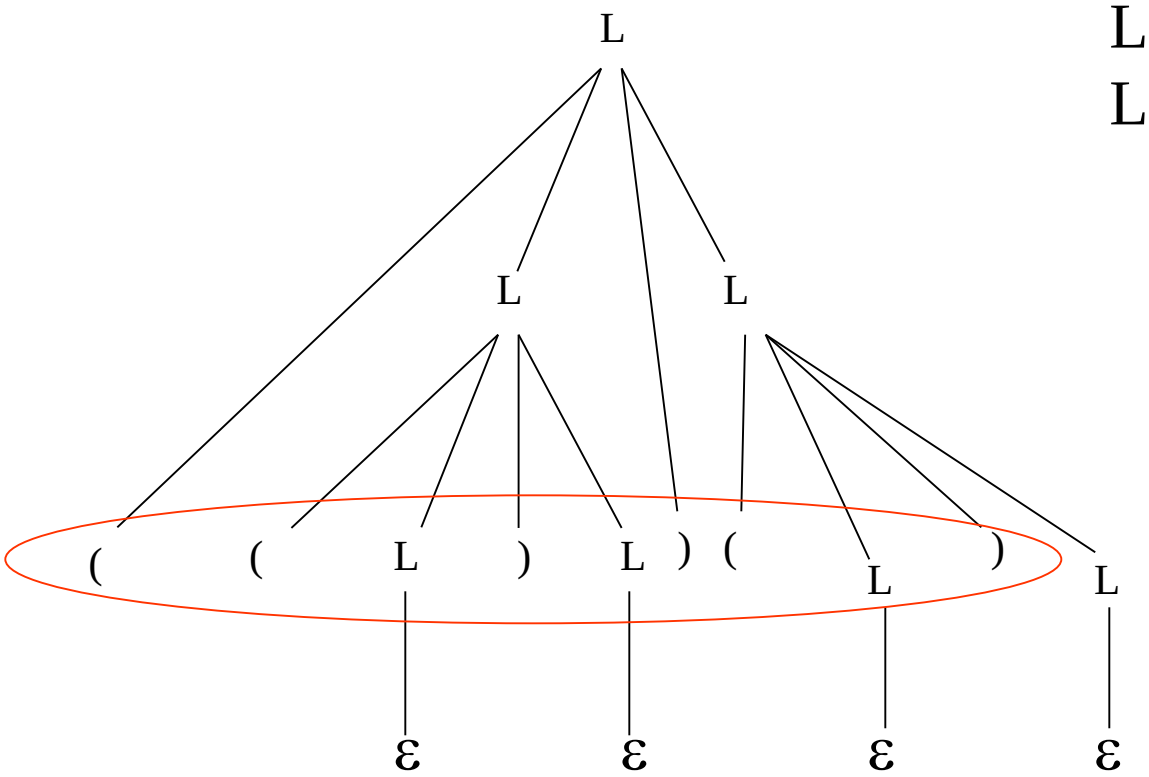
- The task of parsing is figuring out what the parse tree looks like for a given input and language.
- If a string is in the given language, a parse tree must exist.
- However, just because a parse tree exists for some string in a given language doesn’t mean a given algorithm can find it.

Parse Trees

The parse tree for some string in a language that is defined by the grammar G as follows:

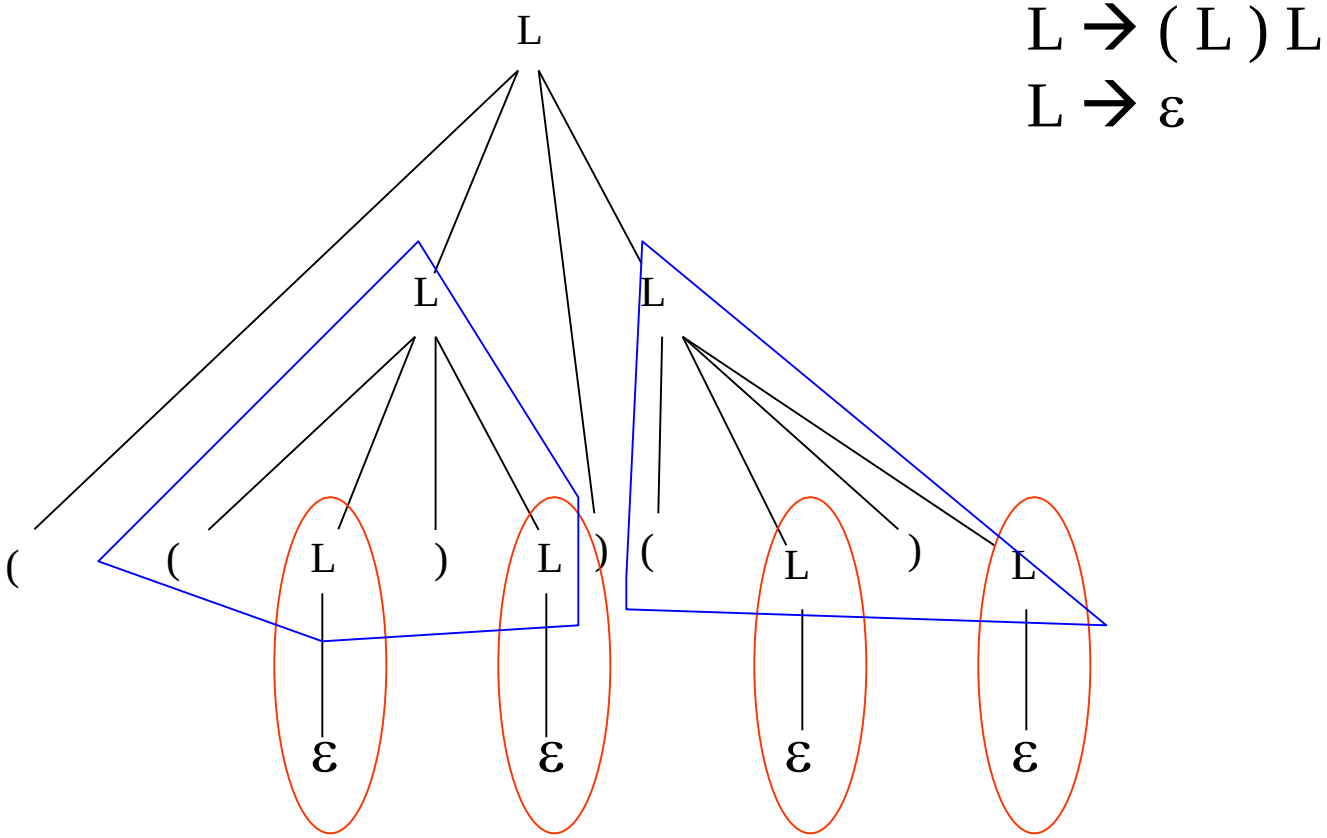
- The root is the start symbol of G
- The leaves are terminals or ε. When visited from left to right, the leaves form the input string
- The interior nodes are non-terminals of G
- For every non-terminal A in the tree with children B_1 … B_k, there is some production A → B_1 … B_k

Parse Tree for (())()



L → (L) L
L → ε

Parse Tree for (())()



Parsing

- Parsing algorithms are based on the idea of *derivations*.
- General Algorithms:
  - LL (top down)
  - LR (bottom up)

Single Step Derivation

Definition: Given $\alpha\ A\ \beta$ (with α,β in $(V_n \cup V_t)^*$) and a production $A \rightarrow \gamma$,

$\alpha\ A\ \beta \Rightarrow \alpha\ \gamma\ \beta$ is a **single step derivation**.

Examples:

$L + D \Rightarrow L - D + D$ $\qquad$ $L \rightarrow L - D$

$(\ L\ )\ (\ L\ ) \Rightarrow (\ (\ L\ )\ L\ )\ (\ L\ )$ $\qquad$ $L \rightarrow (L)\ L$

Greek letters ($\alpha,\beta,\chi,\ldots$) denote a (possibly empty) sequence of terminals and non-terminals.

Derivations

Definition: A sequence of the form:

$w_0 \Rightarrow w_1 \Rightarrow \ldots \Rightarrow w_n$

is a **derivation** of w_n from w_0 ($w_0 \Rightarrow^* w_n$)

L	production L → (L) L
⇒ (L) L	production L → ε
⇒ ( ) L	production L → ε
⇒ ( )	

$L \Rightarrow^* ()$

If w_i has non-terminal symbols, it is referred to as *sentential form*.

$$L \Rightarrow^{*} ((\,))\,(\,)$$

L	production L → (L) L
⇒ (L) **L**	production L → (L) L
⇒ (L) (L) **L**	production L → ε
⇒ (**L**) (L)	production L → (L) L
⇒ ((**L**) L) (L)	production L → ε
⇒ (( ) L) (**L**)	production L → ε
⇒ (( ) **L**) ( )	production L → ε
⇒ (( )) ( )	

- L(G), the language generated by grammar G is {w in V_t^*: s $\Rightarrow^*$ w, for start symbol s}
- We've just shown that both () and (())() are in L(G) for the previous grammar.

Leftmost Derivations

- A derivation where the leftmost nonterminal is always chosen
- If a string is in a given language (i.e. a derivation exists), then a leftmost derivation *must* exist
- Rightmost derivation defined as you would expect

Leftmost Derivation for (())()

L	production L → (L) L
⇒ (**L**) L	production L → (L) L
⇒ ((**L**) L) L	production L → ε
⇒ (( ) **L**) L	production L → ε
⇒ (( )) **L**	production L → ε
⇒ (( )) (**L**) L	production L →(L) L
⇒ (( )) ( ) **L**	production L → ε
⇒ (( )) ( )	

Rightmost Derivation for (())()

L	production L → (L) L
⇒ (L) **L**	production L → (L) L
⇒ (L) (L) **L**	production L → ε
⇒ (L) (**L**)	production L → ε
⇒ (**L**) ( )	production L → (L) L
⇒ ((L) **L**) ( )	production L → ε
⇒ ((**L**)) ( )	production L → ε
⇒ (( )) ( )	

Ambiguity

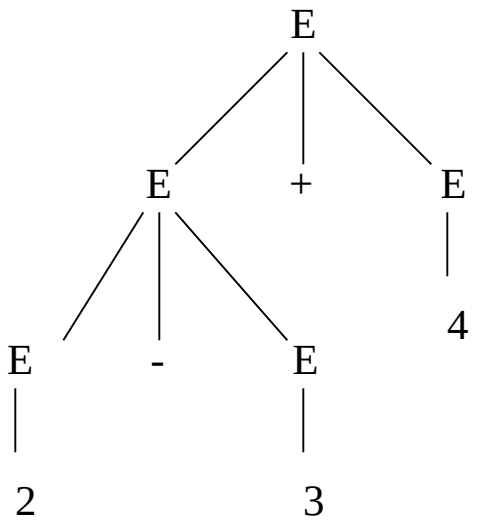
Two (or more) parse trees or leftmost derivations for *some string in the language*

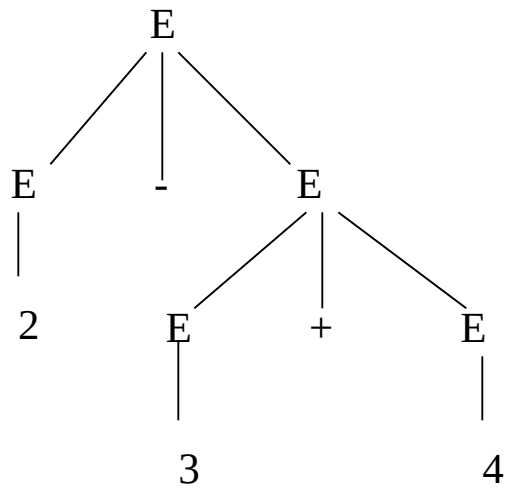
E → E + E

E → E – E

E → 0 | … | 9

2 – 3 + 4





- Two leftmost derivations

$$
\begin{aligned}
E &\Rightarrow E + E \\
&\Rightarrow E - E + E \\
&\Rightarrow 2 - E + E \\
&\Rightarrow 2 - 3 + E \\
&\Rightarrow 2 - 3 + 4
\end{aligned}
\qquad\qquad
\begin{aligned}
E &\Rightarrow E - E \\
&\Rightarrow 2 - E \\
&\Rightarrow 2 - E + E \\
&\Rightarrow 2 - 3 + E \\
&\Rightarrow 2 - 3 + 4
\end{aligned}
$$

- An ambiguous grammar can sometimes be made unambiguous:

  $E \rightarrow E + T \mid E - T \mid T$

  $T \rightarrow 0 \mid \ldots \mid 9$



- Precedence can be specified as well:

  $E \rightarrow E + T \mid E - T \mid T$

  $T \rightarrow T * F \mid T / F \mid F$

  $F \rightarrow (E) \mid 0 \mid \ldots \mid 9$

Input: **`begin simplestmt; simplestmt; end`**

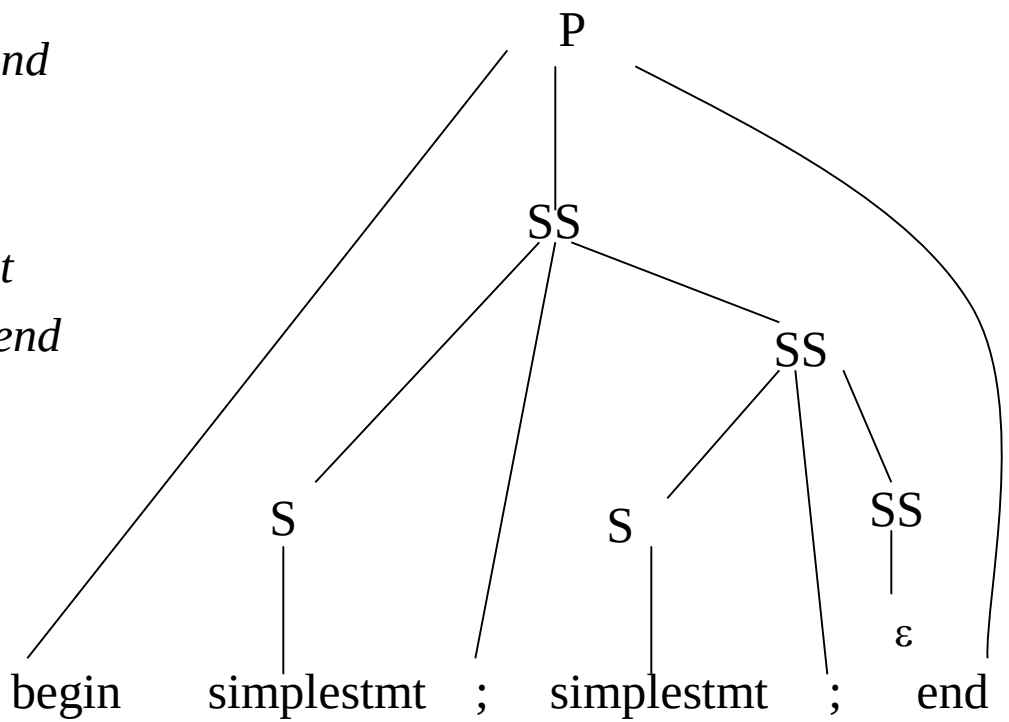
P→ *begin* SS *end*

SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Top Down (LL) Parsing

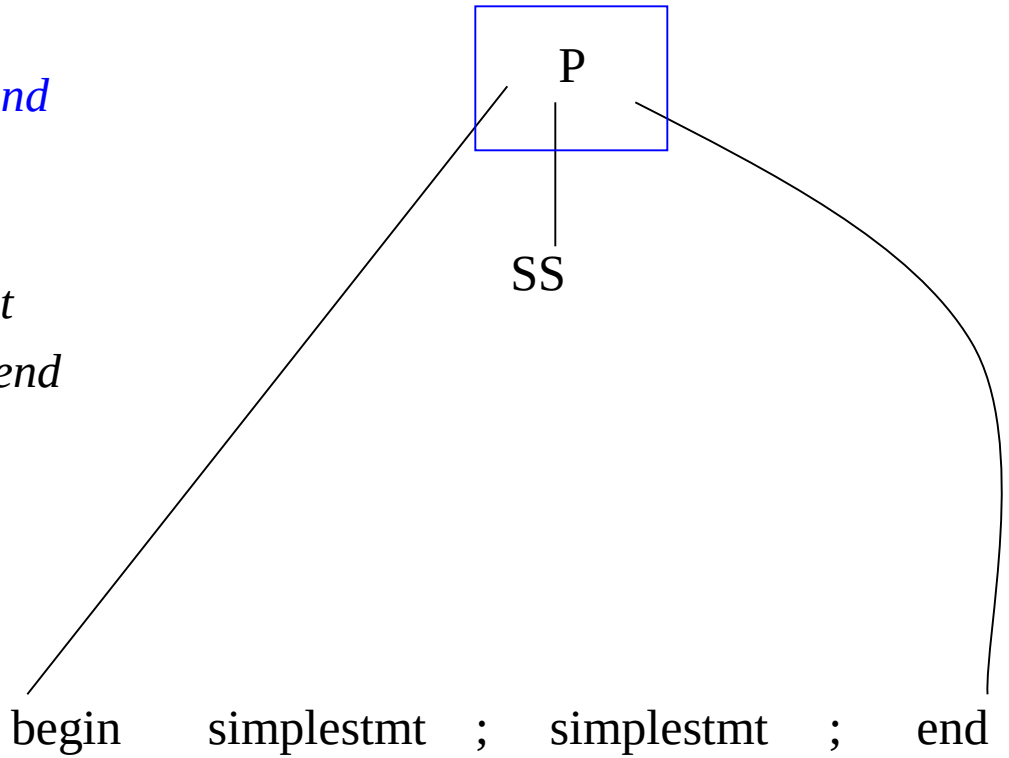
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SS → ε

S → *simplestmt*

S → *begin* SS *end*



Top Down (LL) Parsing

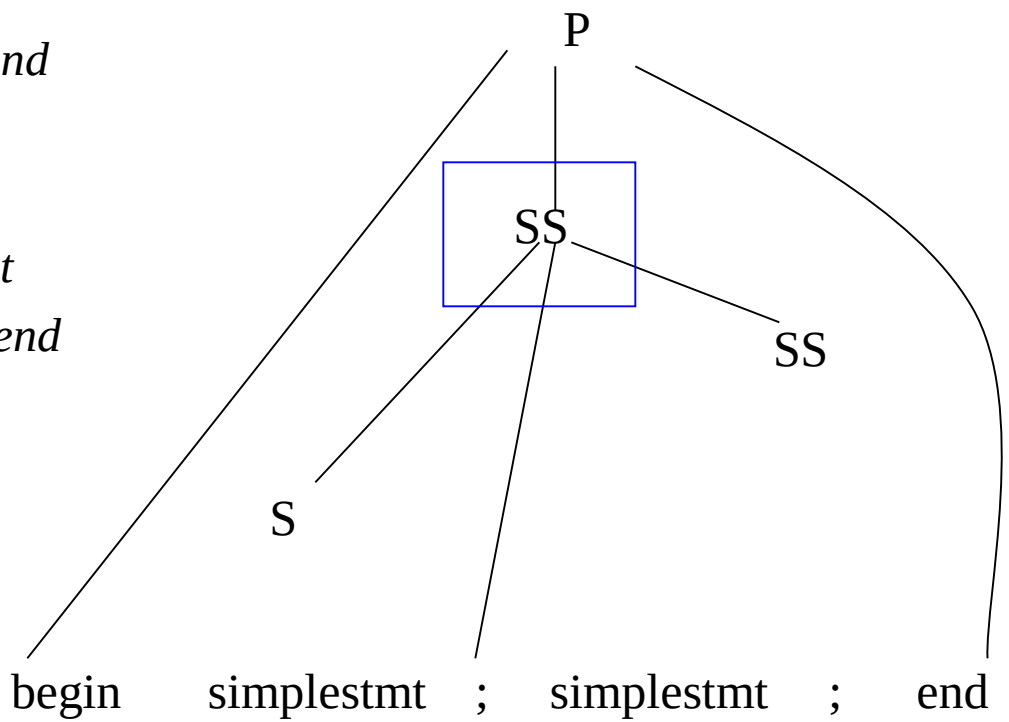
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SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Top Down (LL) Parsing

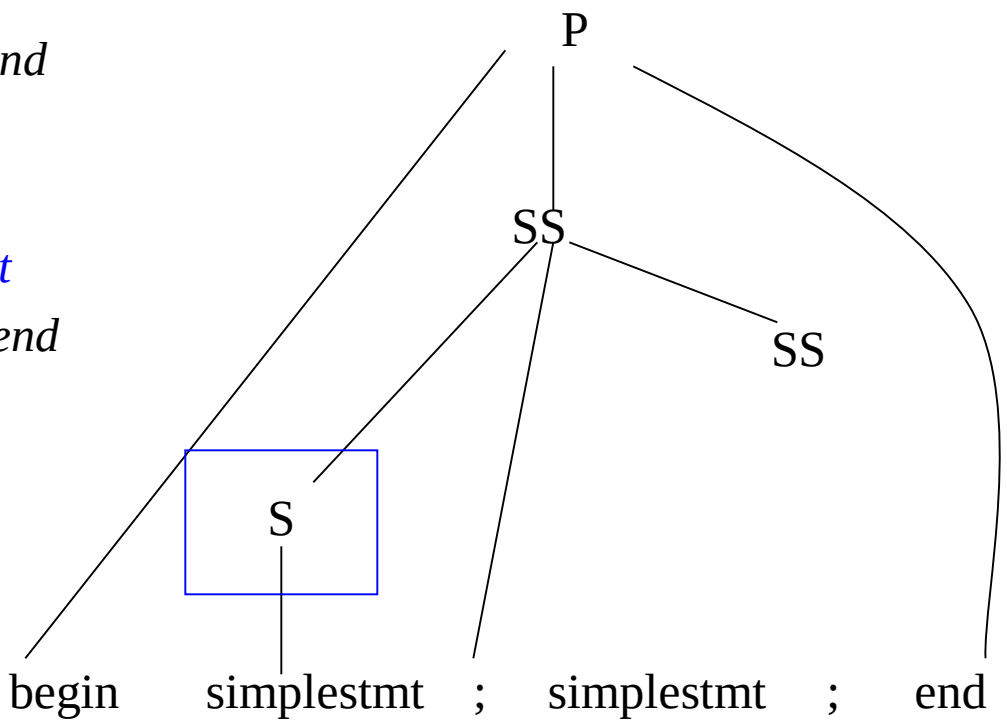
P → *begin* SS *end*

SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Top Down (LL) Parsing

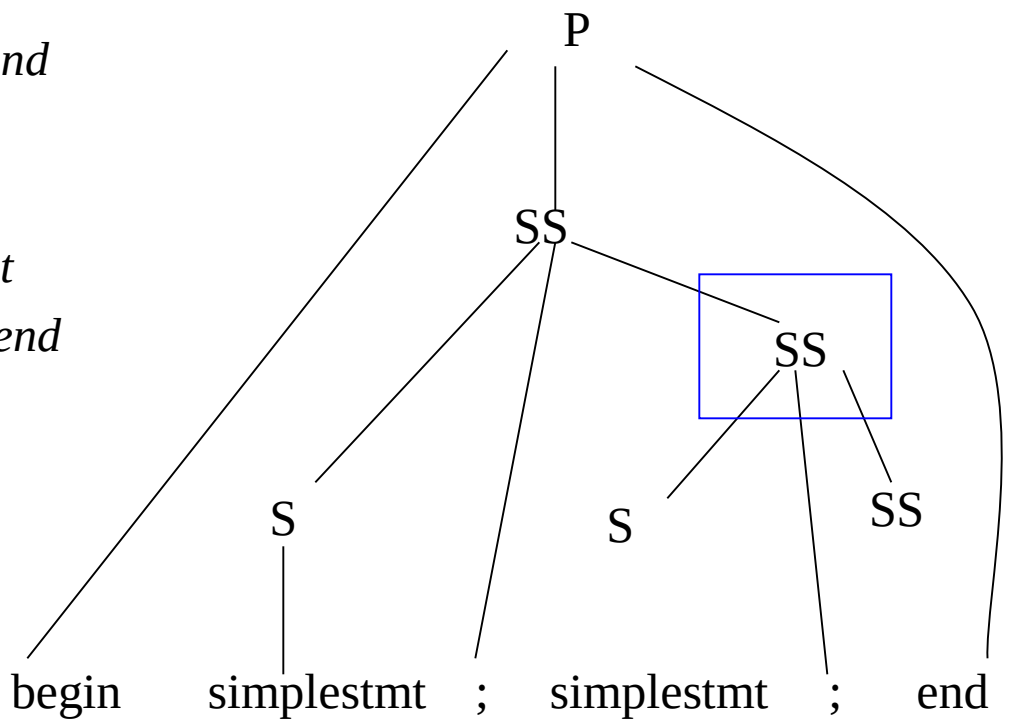
P → *begin* SS *end*

SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Top Down (LL) Parsing

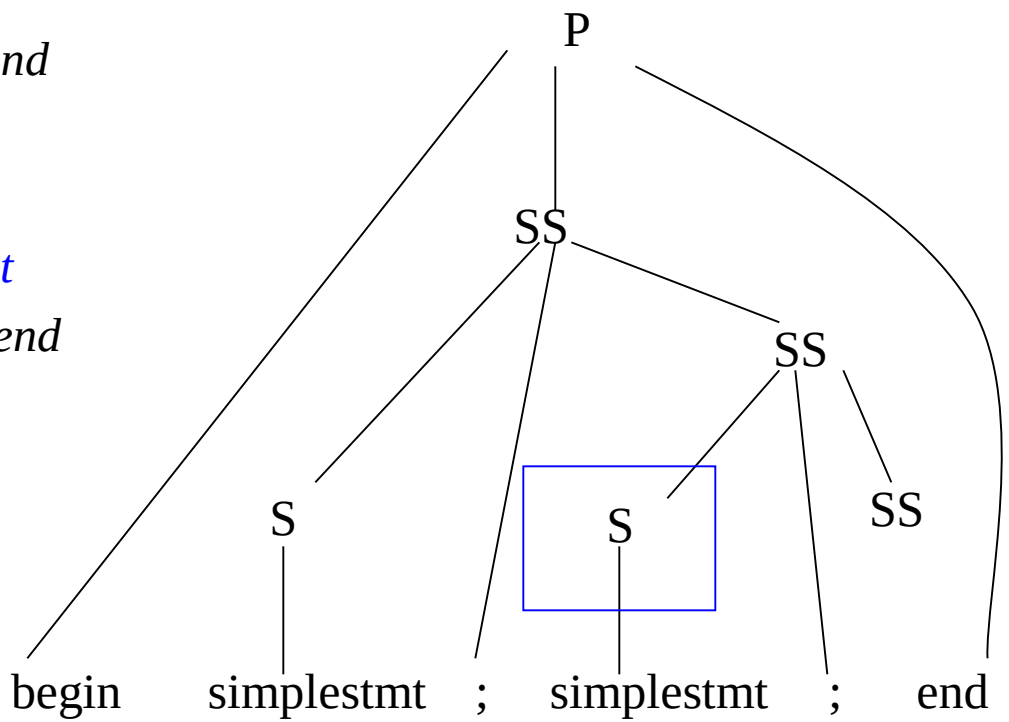
P → *begin* SS *end*

SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Top Down (LL) Parsing

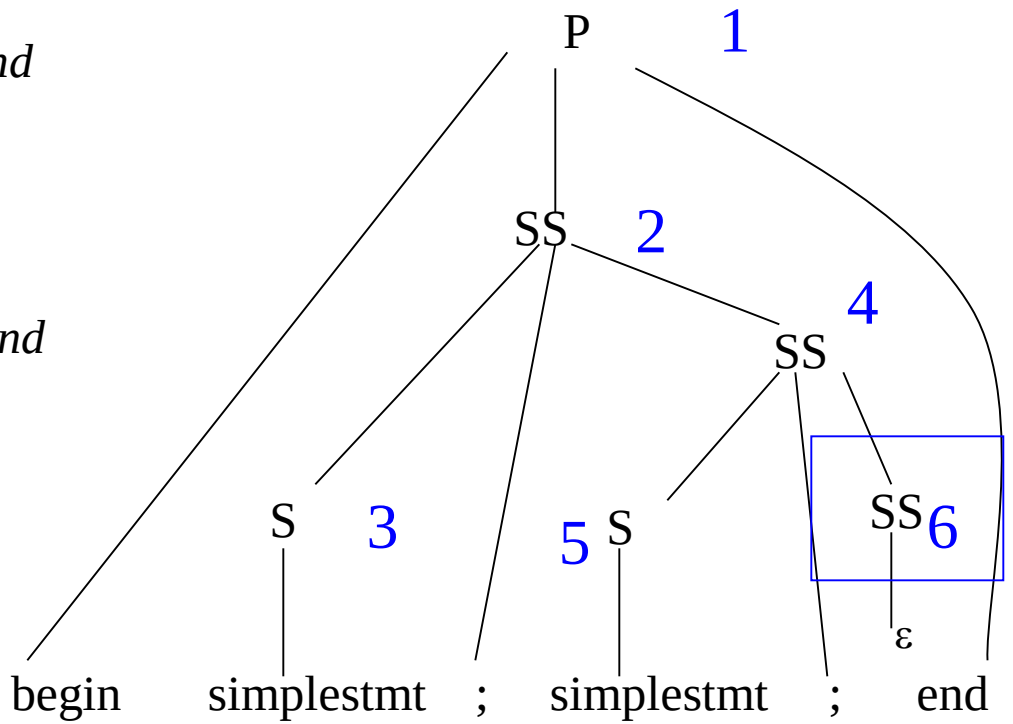
P → *begin* SS *end*

SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Bottomup (LR) Parsing

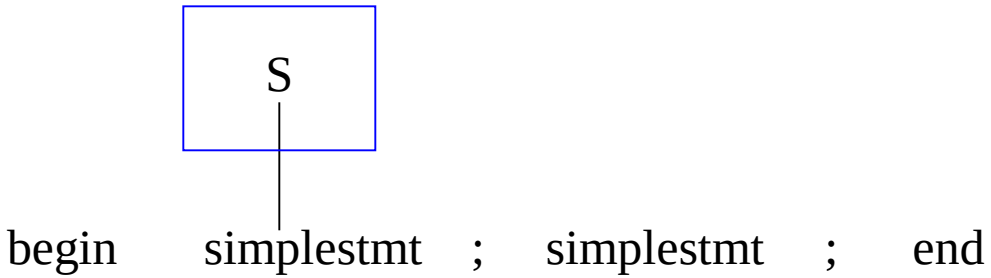
P→ *begin* SS *end*

SS → S *;* SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Bottomup (LR) Parsing

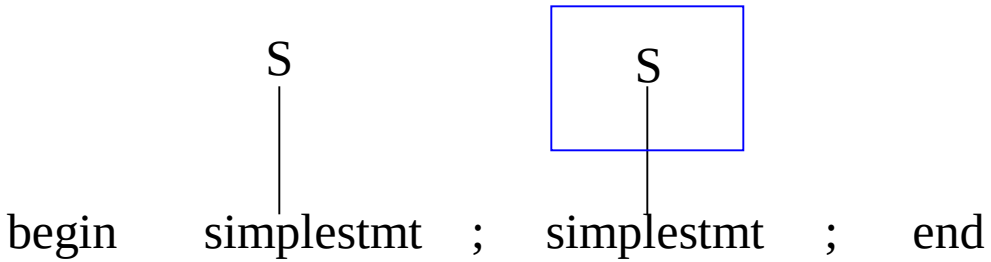
P→ *begin* SS *end*

SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Bottomup (LR) Parsing

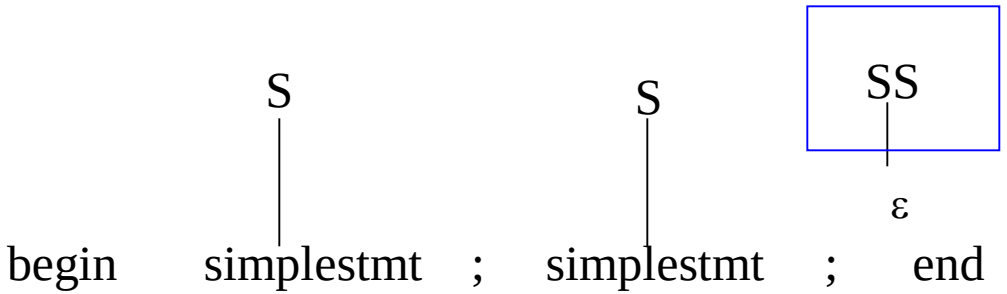
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SS → ε

S → *simplestmt*

S → *begin* SS *end*



Bottomup (LR) Parsing

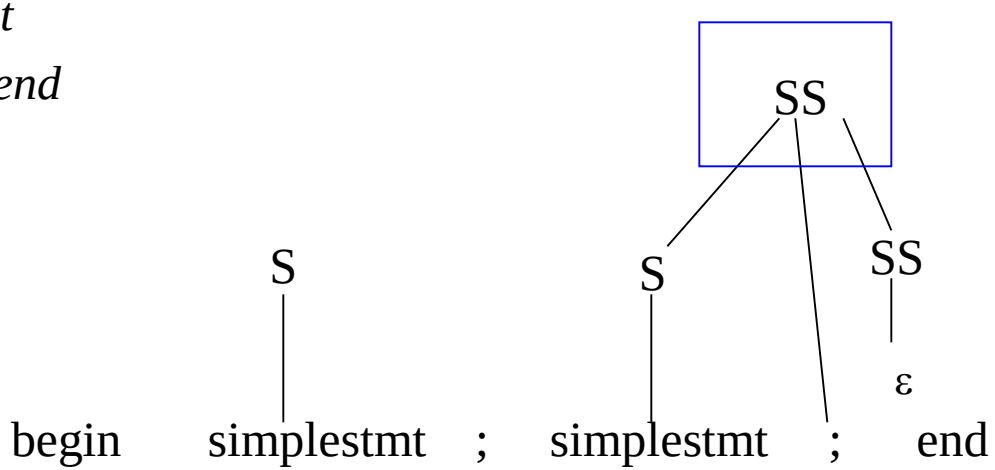
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SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Bottomup (LR) Parsing

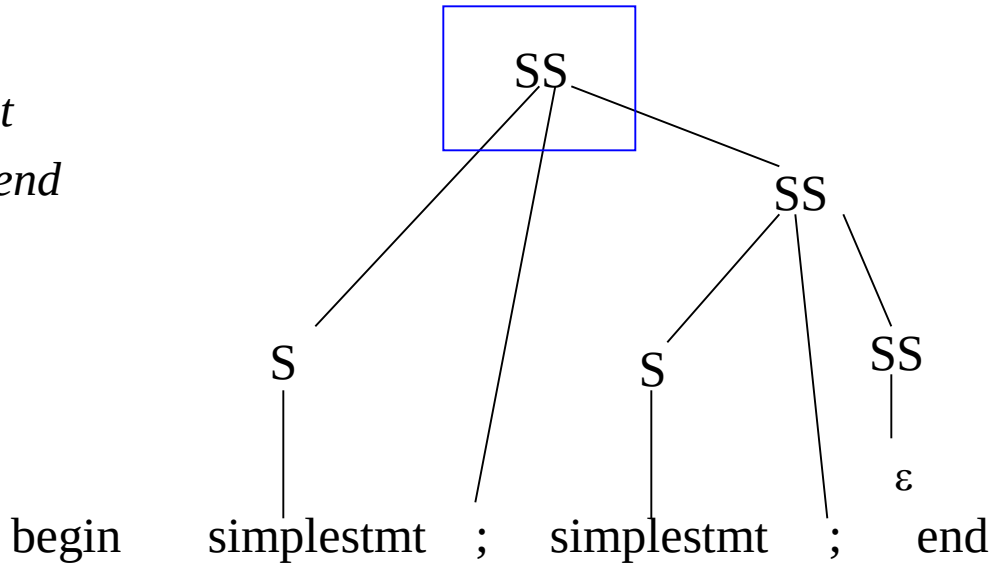
P→ *begin* SS *end*

SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Bottomup (LR) Parsing

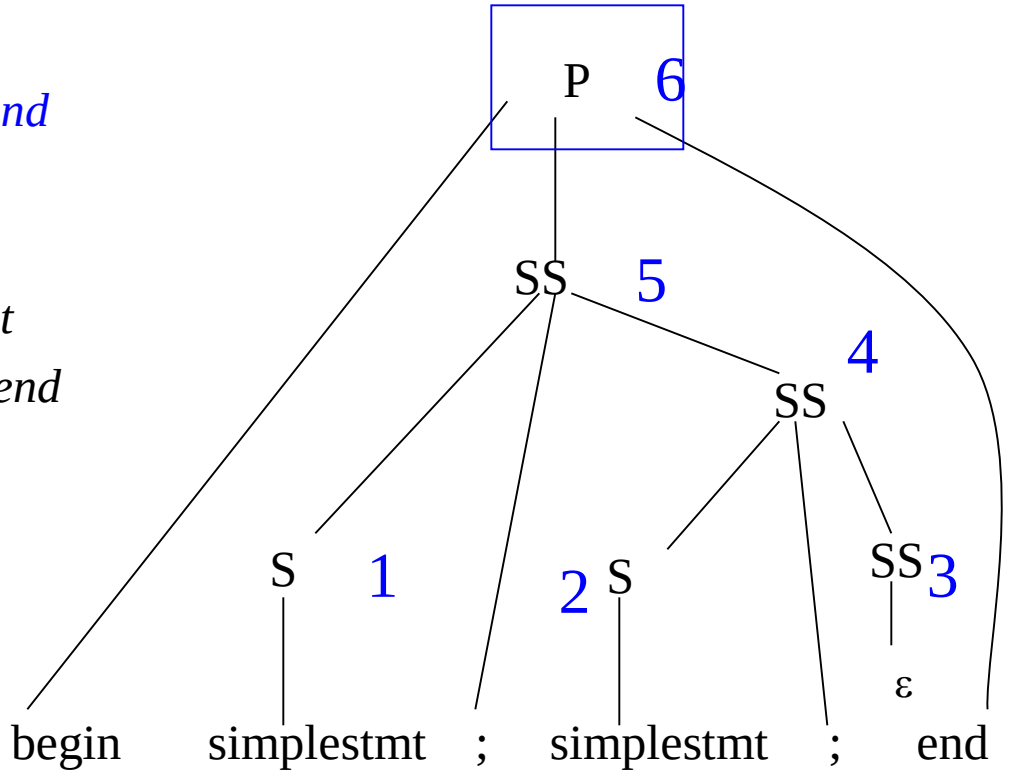
P→ *begin* SS *end*

SS → S ; SS

SS → ε

S → *simplestmt*

S → *begin* SS *end*



Parsing

- General Algorithms:
  - LL (top down)
  - LR (bottom up)
- Both algorithms are driven by the input grammar and the input to be parsed
- Two important sets: FIRST and FOLLOW

FIRST Sets

FIRST(α) is the set of all terminal symbols that can begin some sentential form in a derivation that starts with α

$$\alpha \Rightarrow \ldots \Rightarrow a\,\beta$$

- FIRST(α) = {a in V_t | $\alpha \Rightarrow^* a\beta$ } $\cup$ { ε } if $\alpha \Rightarrow^* \varepsilon$
- Example:

  S → simple | begin S end

  FIRST(S) = {simple, begin}

To compute FIRST across strings of terminals and non-terminals:

$$FIRST(\varepsilon) = \{ \varepsilon \}$$

$$FIRST(A\alpha) = \begin{cases} A & \text{if A is a terminal} \\ FIRST(A) \cup FIRST(\alpha) & \text{if } A \Rightarrow^* \varepsilon \\ FIRST(A) & \text{otherwise} \end{cases}$$

Computing FIRST sets

Initially FIRST(A) (for all A) is empty

1. For productions $A \rightarrow c\ \beta$, where c in V_t

   Add { c } to FIRST(A)

2. For productions $A \rightarrow \varepsilon$

   Add { ε } to FIRST(A)

3. For productions $A \rightarrow \alpha\ B\ \beta$, where $\alpha \Rightarrow^* \varepsilon$ and NOT ($B \Rightarrow^* \varepsilon$)

   Add FIRST(αB) to FIRST(A)

   same thing as $\varepsilon \notin$ FIRST(B)

4. For productions $A \rightarrow \alpha$, where $\alpha \Rightarrow^* \varepsilon$

   Add FIRST(α) and { ε } to FIRST(A)

Example 1

- S → a S e
- S → B
- B → b B e
- B → C
- C → c C e
- C → d

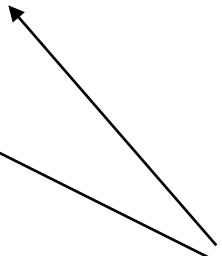
- FIRST(C) = {c,d}
- FIRST(B) =
- FIRST(S) =

Start with the ‘simplest’ non-terminal

Example 1

- S → a S e
- S → B
- B → b B e
- B → C
- C → c C e
- C → d

- FIRST(C) = {c,d}
- FIRST(B) = {b,c,d}
- FIRST(S) =



Now that we know FIRST(C) …

Example 1

- S → a S e
- S → B
- B → b B e
- B → C
- C → c C e
- C → d

- FIRST(C) = {c,d}
- FIRST(B) = {b,c,d}
- FIRST(S) = {a,b,c,d}

Example 2

- P → i | c | n T S
- Q → P | a S | d S c S T
- R → b | ε
- S → e | R n | ε
- T → R S q

- FIRST(P) = {i,c,n}
- FIRST(Q) =
- FIRST(R) = {b,ε}
- FIRST(S) =
- FIRST(T) =

Example 2

- P → i | c | n T S
- Q → P | a S | d S c S T
- R → b | ε
- S → e | R n | ε
- T → R S q

- FIRST(P) = {i,c,n}
- FIRST(Q) = {i,c,n,a,d}
- FIRST(R) = {b,ε}
- FIRST(S) =
- FIRST(T) =

Example 2

- P → i | c | n T S
- Q → P | a S | d S c S T
- R → b | ε
- S → e | R n | ε
- T → R S q

- FIRST(P) = {i,c,n}
- FIRST(Q) = {i,c,n,a,d}
- FIRST(R) = {b,ε}
- FIRST(S) = {e,b,n,ε}
- FIRST(T) =

Note:

$S \Rightarrow R\ n \Rightarrow n$ because $R \Rightarrow^* \varepsilon$

Example 2

- P → i | c | n T S
- Q → P | a S | d S c S T
- R → b | ε
- S → e | R n | ε
- T → R S q

- FIRST(P) = {i,c,n}
- FIRST(Q) = {i,c,n,a,d}
- FIRST(R) = {b,ε}
- FIRST(S) = {e,b,n,ε}
- FIRST(T) = {b,c,n,q}

Note:

$T \Rightarrow R\ S\ q \Rightarrow S\ q \Rightarrow q$

because both R and S $\Rightarrow^*$ ε

Example 3

- S → a S e | S T S
- T → R S e | Q
- R → r S r | ε
- Q → S T | ε

- FIRST(S) =
- FIRST(R) =
- FIRST(T) =
- FIRST(Q) =

Example 3

- S → a S e | S T S
- T → R S e | Q
- R → r S r | ε
- Q → S T | ε

- FIRST(S) = {a}
- FIRST(R) = {r, ε}
- FIRST(T) = {r,a, ε}
- FIRST(Q) = {a, ε}

FOLLOW Sets

- FOLLOW(A) is the set of terminals (including end of file - $) that may follow non-terminal A in some sentential form.
- FOLLOW(A) = {c in V_t | $S \Rightarrow^+ \ldots Ac \ldots\} \cup$ {$} if $S \Rightarrow^+ \ldots A$
- For example, consider $L \Rightarrow^+ (())(L)L$

  Both ‘)’ and end of file can follow L
- NOTE: ε is ***never*** in FOLLOW sets

Computing FOLLOW(A)

1. If A is start symbol, put $ in FOLLOW(A)
2. Productions of the form $B \rightarrow \alpha A \beta$,

   Add FIRST(β) – {ε} to FOLLOW(A)

INTUITION: Suppose $B \rightarrow AX$ and FIRST(X) = {c}

$S \Rightarrow^{+} \alpha B \beta \Rightarrow \alpha A X \beta \Rightarrow^{+} \alpha A c \delta \beta$

= FIRST(X)

3. Productions of the form $B \rightarrow \alpha A$ or

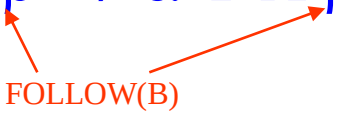
$B \rightarrow \alpha A \beta$ where $\beta \Rightarrow^* \varepsilon$

Add FOLLOW(B) to FOLLOW(A)

INTUITION:

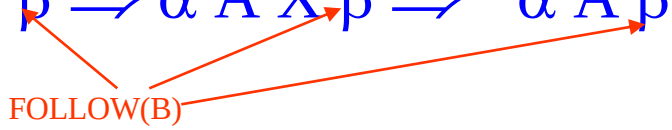
- Suppose $B \rightarrow Y A$

$S \Rightarrow^+ \alpha B \beta \Rightarrow \alpha Y A \beta$



- Suppose $B \rightarrow A X$ and $X \Rightarrow^* \varepsilon$

$S \Rightarrow^+ \alpha B \beta \Rightarrow \alpha A X \beta \Rightarrow^* \alpha A \beta$



Assume the first non-terminal is the start symbol

Example 4

- S → a S e | B
- B → b B C f | C
- C → c C g | d | ε

- FIRST(C) = {c,d,ε}
- FIRST(B) = {b,c,d,ε}
- FIRST(S) = {a,b,c,d,ε}

- FOLLOW(C) =
- FOLLOW(B) =
- FOLLOW(S) = {$}

Using rule #1

Example 4

- S → a S e | B
- B → b B C f | C
- C → c C g | d | ε

- FIRST(C) = {c,d,ε}
- FIRST(B) = {b,c,d,ε}
- FIRST(S) = {a,b,c,d,ε}

- FOLLOW(C) = {f,g}
- FOLLOW(B) = {c,d,f}
- FOLLOW(S) = {$,e}

Using rule #2

Example 4

- S → a S e | B
- B → b B C f | C
- C → c C g | d | ε

- FIRST(C) = {c,d,ε}
- FIRST(B) = {b,c,d,ε}
- FIRST(S) = {a,b,c,d,ε}

- FOLLOW(C) = {f,g} ∪ FOLLOW(B) = {c,d,e,f,g,$}
- FOLLOW(B) = {c,d,f} ∪ FOLLOW(S) = {c,d,e,f,$}
- FOLLOW(S) = {$, e }

Using rule #3

Example 5

- S → A B C | A D
- A → ε | a A
- B → b | c | ε
- C → D d C
- D → e b | f c

- FIRST(D) = {e,f}
- FIRST(C) = {e,f}
- FIRST(B) = {b,c,ε}
- FIRST(A) = {a, ε}
- FIRST(S) = {a,b,c,e,f}

- FOLLOW(S) =
- FOLLOW(A) =
- FOLLOW(B) =
- FOLLOW(C) =
- FOLLOW(D) =

Example 5

- S → A B C | A D
- A → ε | a A
- B → b | c | ε
- C → D d C
- D → e b | f c

- FIRST(D) = {e,f}
- FIRST(C) = {e,f}
- FIRST(B) = {b,c,ε}
- FIRST(A) = {a, ε}
- FIRST(S) = {a,b,c,e,f}

- FOLLOW(S) = {$}
- FOLLOW(A) = {b,c,e,f}
- FOLLOW(B) = {e,f}
- FOLLOW(C) = {$}
- FOLLOW(D) = {$}

Example 6

- S → (A) | ε
- A → T E
- E → & T E | ε
- T → (A) | a | b | c

- FIRST(T) = {(,a,b,c}
- FIRST(E) = {&, ε }
- FIRST(A) = {(,a,b,c}
- FIRST(S) = {(, ε}

- FOLLOW(S) =
- FOLLOW(A) =
- FOLLOW(E) =
- FOLLOW(T) =

Example 6

- S → (A) | ε
- A → T E
- E → & T E | ε
- T → (A) | a | b | c

- FIRST(T) = {(,a,b,c}
- FIRST(E) = {&, ε }
- FIRST(A) = {(,a,b,c}
- FIRST(S) = {(, ε}

- FOLLOW(S) = {$}
- FOLLOW(A) = {) }
- FOLLOW(E) =

FOLLOW(A) = {) }

- FOLLOW(T) =

FIRST(E) ∪ FOLLOW(A) ∪ FOLLOW(E) = {&,)}

Example 7

- E → T E’
- E’ → + T E’ | ε
- T → F T’
- T’ → * F T’ | ε
- F → (E) | id

- FOLLOW(E) =
- FOLLOW(E’) =
- FOLLOW(T) =
- FOLLOW(T’) =
- FOLLOW(F) =

- FIRST(F) = FIRST(T) = FIRST(E) = {(,id}
- FIRST(T’) = {*,ε}
- FIRST(E’) = {+,ε}

Example 7

- E → T E’
- E’ → + T E’ | ε
- T → F T’
- T’ → * F T’ | ε
- F → (E) | id

- FIRST(F) = FIRST(T) = FIRST(E) = {(,id}
- FIRST(T’) = {*,ε}
- FIRST(E’) = {+,ε}

- FOLLOW(E) = {$,)}
- FOLLOW(E’) = FOLLOW(E) = {$,)}
- FOLLOW(T) = FIRST(E’) ∪ FOLLOW(E) ∪ FOLLOW(E’) = {+,$,)}
- FOLLOW(T’) = FOLLOW(T) = {+,$,)}
- FOLLOW(F) = FIRST(T’) ∪ FOLLOW(T) ∪ FOLLOW(T’) = {*,+,$,)}